

C673

General Chemistry II with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A rigid tank painted green, delivered to the lab on Monday, holds a gas at 2.0 atm and 27°C. It is warmed to 127°C with no change in volume. What is the new pressure?

- A. About 9.4 atm
- B. 1.5 atm
- C. About 2.7 atm
- D. 2.0 atm, since the tank is sealed

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

At constant pressure, a gas at 300 K occupies 4.0 L. To what temperature must it be heated to reach 6.0 L?

- A. 177 K
- B. 302 K
- C. 200 K
- D. 450 K

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A teacher who has prepared solutions for this lab for eight years adds 250 mL of distilled water to 50 mL of 6.0 M HCl. The bottle's label is slightly faded. What is the new concentration?

- A. 1.0 M
- B. 36 M
- C. 6.0 M, because the moles of HCl do not change
- D. 1.2 M

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

For the reaction $2A + B \rightarrow C$, run in a lab that just got new stir plates, experiments show rate = $k[A]^2[B]$. If $[A]$ is doubled and $[B]$ is held constant, how does the rate change?

- A. It increases by a factor of 4.
- B. It increases by a factor of 8.
- C. It doubles.
- D. It is cut to one-fourth.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

When the concentration of reactant X is tripled with everything else held constant, the rate increases by a factor of 9. What is the order with respect to X?

- A. 9
- B. 1
- C. 2
- D. 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A fertilizer plant that employs 400 people makes ammonia: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta H = -92 \text{ kJ}$. What happens to the equilibrium if the pressure is increased by compressing the mixture?

- A. It shifts right because the reaction releases heat.
- B. It shifts right, toward fewer gas moles.
- C. It does not shift, because K never changes.
- D. It shifts left.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a biochemistry lab that uses enzymes from a supplier in Wisconsin, a student adds catalase to hydrogen peroxide and sees rapid bubbling. The student used a 10 mL graduated cylinder. What effect does the enzyme have?

- A. It raises the activation energy so more molecules react.
- B. It lowers the activation energy, speeding the reaction without changing ΔH .
- C. It is used up, so more must be added for every molecule.
- D. It increases the amount of product at equilibrium.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A water-quality teacher whose students test a creek each spring prepares 0.001 M HCl, a strong acid, in a 1-liter bottle. What is its pH?

- A. 0.001
- B. 11
- C. -3
- D. 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A solution has a pH of 4 at 25°C. What is $[\text{OH}^-]$?

- A. $1 \times 10^{-10} \text{ M}$
- B. 10 M
- C. $1 \times 10^{-4} \text{ M}$
- D. $1 \times 10^4 \text{ M}$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher who runs a battery-building club places a zinc strip in copper(II) sulfate solution: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$. The solution fades from blue over 20 minutes. Which species is the oxidizing agent?

- A. Cu
- B. Cu^{2+}
- C. Zn^{2+}
- D. Zn

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A hospital that serves three counties receives 80 mg of a tracer with a half-life of 5 days. The shipment arrived on a Tuesday. How much remains after 15 days?

- A.** None, since it is fully used up by then
- B.** 20 mg
- C.** 10 mg
- D.** About 26.7 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A 100 g sample decays to 12.5 g in 24 hours. What is the half-life?

- A.** 24 hours
- B.** 12 hours
- C.** 3 hours
- D.** 8 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A rigid tank ~~painted green, delivered to the lab on Monday,~~ holds a gas at 2.0 atm and 27°C. It is warmed to 127°C with no change in volume. What is the new pressure?

Target word: new pressure **Decides:** Convert to kelvin: 300 K to 400 K.

A	Dropped element	Uses Celsius: $2.0 \times 127/27$.
B	Inverted operation	Multiplies by 300/400, as if pressure fell with heating.
C	Correct	$2.0 \times 400/300 = 2.67$ atm.
D	Confusion set	Fixed volume, not fixed pressure; pressure rises with temperature.

2. Answer D Concept 1 • transfer

At constant pressure, a gas at 300 K occupies 4.0 L. To what temperature must it be heated to reach 6.0 L?

Target word: solve for T_2 **Decides:** $T_2 = T_1 \times V_2/V_1$.

A	Confusion set	Gives the Celsius value with a kelvin label.
B	Adjacent category	Adds the volume change to the temperature.
C	Inverted operation	Multiplies by V_1/V_2 .
D	Correct	$300 \times 6.0/4.0 = 450$ K (177°C).

3. Answer A Concept 2

A teacher ~~who has prepared solutions for this lab for eight years~~ adds 250 mL of distilled water to 50 mL of 6.0 M HCl. ~~The bottle's label is slightly faded.~~ What is the new concentration?

Target word: diluted molarity **Decides:** Final volume = 50 + 250 = 300 mL.

A	Correct	$6.0 \times 50 / 300 = 1.0$ M.
B	Inverted operation	Multiplies by 300/50.
C	Confusion set	Moles stay fixed; concentration falls as volume grows.
D	Dropped element	Uses 250 mL of water as the final volume.

4. Answer A Concept 3

For the reaction ~~$2A + B \rightarrow C$, run in a lab that just got new stir plates,~~ experiments show rate = $k[A]^2[B]$. If [A] is doubled and [B] is held constant, how does the rate change?

Target word: rate change **Decides:** Second order in A: $2^2 = 4$.

A	Correct	Doubling a second-order reactant multiplies the rate by 4.
B	Adjacent category	That would be doubling both A and B.
C	Dropped element	Ignores the exponent 2.
D	Inverted operation	Higher concentration speeds, not slows, this reaction.

5. Answer C Concept 3 • transfer

When the concentration of reactant X is tripled with everything else held constant, the rate increases by a factor of 9. What is the order with respect to X?

Target word: solve for the order **Decides:** $3^m = 9$, so $m = 2$.

A	Confusion set	The rate factor, not the order.
B	Adjacent category	First order would only triple the rate.
C	Correct	$3^2 = 9$.
D	Wrong stem number	The concentration factor, not the order.

6. Answer B Concept 4

A fertilizer plant ~~that employs 400 people~~ makes ammonia: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $\Delta H = -92 \text{ kJ}$. What happens to the equilibrium if the pressure is increased by compressing the mixture?

Target word: direction of shift under pressure **Decides:** 4 moles of gas on the left, 2 on the right.

A	Camouflage	ΔH predicts temperature effects, not pressure effects.
B	Correct	Higher pressure favors the side with fewer gas particles.
C	Absolute language	K is fixed at constant temperature, but the position still shifts to restore $Q = K$.
D	Inverted operation	Left has more gas moles.

7. Answer B Concept 5

In a biochemistry lab ~~that uses enzymes from a supplier in Wisconsin~~, a student adds catalase to hydrogen peroxide and sees rapid bubbling. ~~The student used a 10 mL graduated cylinder.~~ What effect does the enzyme have?

Target word: catalyst effect **Decides:** Enzymes are catalysts.

A	Inverted operation	A higher barrier would slow the reaction.
B	Correct	That is what any catalyst does.
C	Adjacent category	Catalysts are regenerated and reused.
D	Confusion set	Catalysts change rate, not equilibrium yield.

8. Answer D Concept 6

A water-quality teacher ~~whose students test a creek each spring~~ prepares 0.001 M HCl, a strong acid, ~~in a 1-liter bottle~~. What is its pH?

Target word: pH **Decides:** $[\text{H}^+] = 1 \times 10^{-3} \text{ M}$.

A	Wrong stem number	The concentration, not the pH.
B	Inverted operation	That is the pOH.
C	Dropped element	Leaves out the negative sign in $-\log$.
D	Correct	$-\log(10^{-3}) = 3$.

9. Answer A Concept 6 • transfer

A solution has a pH of 4 at 25°C. What is $[\text{OH}^-]$?

Target word: solve for hydroxide **Decides:** $\text{pOH} = 14 - 4 = 10$.

A	Correct	$[\text{OH}^-] = 10^{-\text{pOH}}$.
B	Dropped element	Reports pOH as a concentration.
C	Confusion set	That is $[\text{H}^+]$.
D	Inverted operation	Sign error on the exponent.

10. Answer B Concept 7

A teacher ~~who runs a battery building club~~ places a zinc strip in copper(II) sulfate solution: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$. The solution fades from blue ~~over 20 minutes~~. Which species is the oxidizing agent?

Target word: oxidizing agent **Decides:** Cu^{2+} gains electrons (is reduced).

A	Confusion set	The product of the reduction, not the agent.
B	Correct	The species reduced is the oxidizing agent.
C	Adjacent category	A product, not an agent.
D	Inverted operation	Zn is oxidized, so it is the reducing agent.

11. Answer C Concept 8

A hospital ~~that serves three counties~~ receives 80 mg of a tracer with a half-life of 5 days. ~~The shipment arrived on a Tuesday.~~ How much remains after 15 days?

Target word: amount remaining **Decides:** 15 days = 3 half-lives.

A	Absolute language	Halving never reaches zero in a set number of steps.
B	Dropped element	Counts only two half-lives.
C	Correct	$80 \rightarrow 40 \rightarrow 20 \rightarrow 10$ mg.
D	Confusion set	Divides by 3 instead of halving three times.

12. Answer D Concept 8 • transfer

A 100 g sample decays to 12.5 g in 24 hours. What is the half-life?

Target word: solve for half-life **Decides:** $100 \rightarrow 12.5$ is three halvings.

A	Wrong stem number	The total time given.
B	Dropped element	Treats the decay as two halvings.
C	Confusion set	That is the number of half-lives.
D	Correct	$24 \text{ hours} / 3 \text{ half-lives} = 8 \text{ hours}$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Gas laws use kelvin	$P_1V_1/T_1 = P_2V_2/T_2$ holds only with absolute temperature ($K = ^\circ C + 273$).	Plugging Celsius temperatures into the ratio.	Temperatures given in $^\circ C$ in a gas problem.
2	Dilution	$M_1V_1 = M_2V_2$, where V_2 is the final total volume; moles of solute stay the same.	Using the volume of water added as the final volume.	"Water is added to" a stock solution.
3	Rate laws and reaction order	$\text{Rate} = k[A]^m[B]^n$; orders come from experiments, not from the balanced equation's coefficients.	Reading exponents from the coefficients, or assuming rate scales linearly.	"Doubling the concentration of ..." with a rate law or data table.
4	Le Chatelier's principle	A system at equilibrium shifts to counteract a change; raising pressure favors the side with fewer gas moles.	Using ΔH to predict a pressure change, or expecting no shift.	Gas-phase equilibrium with a change in pressure, concentration, or temperature.
5	Catalysts and enzymes	A catalyst lowers activation energy and speeds forward and reverse reactions equally; it does not change ΔH , K , or the equilibrium amounts.	Thinking a catalyst increases product yield at equilibrium.	An enzyme or catalyst added to a reaction.
6	pH and pOH	$\text{pH} = -\log[H^+]$; $\text{pH} + \text{pOH} = 14$ at $25^\circ C$; each pH unit is a tenfold change.	Treating pH as linear or mixing up $[H^+]$ and $[OH^-]$.	A concentration of a strong acid or base, or a pH value.
7	Oxidation and reduction	Oxidation is loss of electrons; the substance reduced is the oxidizing agent, and the substance oxidized is the reducing agent.	Naming the oxidized species as the oxidizing agent.	A metal replacing a metal ion, or "oxidizing agent."
8	Radioactive half-life	After each half-life, half the remaining sample decays: $\text{amount left} = \text{start} \times (1/2)^{(t / \text{half-life})}$.	Subtracting a fixed amount each half-life, or expecting the sample to be gone after two half-lives.	A starting mass, a half-life, and an elapsed time.

Course scope (public description): General Chemistry II with Lab for graduates continues the study of general chemistry for already-licensed teachers seeking an additional license or endorsement in secondary chemistry. Building on the topics covered in General Chemistry I, General Chemistry II examines the behavior of gases and solutions, reaction rates and equilibrium, acids and bases, and oxidation-reduction reactions. Also, this course provides an introduction to three subdisciplines of chemistry: organic chemistry, biochemistry, and nuclear chemistry. Laboratory experiences reinforce the essential skills required for conducting successful scientific investigations. C672: General Chemistry I for graduates is a prerequisite for this course.